

End-of-Studies Internship Report

Program: Artificial Intelligence and Computer Engineering

• • Safae Nagbi • •

Title

*Agricultural Intelligence:
Crop Mapping using Machine Learning
and Sentinel-2 Satellite Imagery*

Host Company: Oracle Morocco Research and Development Center

Professional Supervisor: Mitra Noosha

Academic Supervisor: Soukaina Bouhsissin

From 15/01/2026 **to** 14/07/2026

Academic Year: 2025–2026

*To my beloved family,
for their unconditional love, endless patience,
and constant encouragement throughout my
journey.*

*To my dear friends,
for their support, motivation, and presence
during both the challenging and joyful moments.*

*This work is dedicated to all those
who believed in me and helped me move forward.*

— Safae Nagbi

Acknowledgments

I would like to express my sincere gratitude to all those who contributed, directly or indirectly, to the successful completion of this internship and the preparation of this report.

First, I would like to thank Oracle Corporation and the Oracle Morocco Research and Development Center for welcoming me and giving me the opportunity to carry out this internship in a professional and stimulating environment. This experience allowed me to strengthen my technical skills, discover real-world industrial practices, and better understand the challenges related to digital government and data-driven solutions.

I would like to express my special thanks to my internship mentor, Mitra Noosha, for her guidance, availability, valuable advice, and continuous support throughout the project. Her supervision helped me better understand the project objectives, overcome technical challenges, and progress with confidence during the different phases of the internship.

I am also deeply grateful to the members of my team, especially Sam Batterman, ElQajjam Mohammed, Hamza EL MAKRINI, Lahoucine Digrari, Anass Zouaiga, Sara Essalhi, Imane Jabal, and all the other team members, for their warm welcome, collaboration, support, and willingness to share their knowledge and experience.

I would like to thank my academic mentor, Soukaina Bouhsissin, as well as the ENSAM pedagogical team and all my professors for their guidance, support, and the knowledge they have provided throughout my academic journey.

Finally, I would like to express my deepest appreciation to my family and friends for their constant encouragement, patience, and support. Their presence and motivation have been invaluable throughout this experience.



Résumé

Résumé : Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Mots-clés : mot1, mot2, mot3, mot4, mot5.

Summary

Abstract: Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

Keywords: keyword1, keyword2, keyword3, keyword4, keyword5.

List of Figures

1.1	Oracle MADC Location	3
1.2	MADC Major Milestones	3
2.1	Jira Ticket Example	11
2.2	Generic Training	11
2.3	Agile Methodology	14
2.4	Gantt Chart	14
3.1	Sample figure in Chapter 3	19

List of Tables

3.1	Sample table in Chapter 3	19
-----	-------------------------------------	----

List of Listings

3.1	Sample Python listing	19
-----	---------------------------------	----

Contents

Dedication	ii
Acknowledgments	iii
Résumé	iv
Summary	v
List of Figures	vi
List of Tables	vii
List of Listings	viii
1 General Context	2
1.1 Introduction	2
1.2 Presentation of the Host Organization	2
1.2.1 Oracle Corporation	2
1.2.2 Oracle Morocco R&D Center	2
1.2.3 Business Area of Activity	4
1.2.3.1 Oracle Morocco R&D Center Lines of Businesses (LOBs)	4
1.2.3.1.1 Oracle APEX	4
1.2.3.1.2 Oracle Health and Artificial Intelligence	4
1.2.3.1.3 Oracle Linux	4
1.2.3.1.4 Oracle NetSuite	4
1.2.3.1.5 Oracle Database Development Tools	5
1.2.3.1.6 Fusion ERP	5
1.2.3.1.7 Oracle Analytics	5
1.2.3.1.8 Oracle Cloud Infrastructure	5
1.2.3.1.9 Oracle Digital Government Group	5
1.2.3.1.10 Oracle Java Platform Group	5
1.2.4 Organizational Structure	5
1.2.5 Organizational Culture	6
1.3 Topic Context	7
1.3.1 General Context of the Project	7

1.3.2	Problematic	7
1.3.3	Solution	8
1.4	Conclusion	9
2	Project Management	10
2.1	Introduction	10
2.2	Internship Process	10
2.2.1	Onboarding Phase	10
2.2.2	Training Phase	10
2.3	Management Tools	11
2.3.1	Confluence	11
2.3.2	Zoom	12
2.3.3	Jira	12
2.3.4	Slack	12
2.4	Project Methodology	12
2.5	Gantt Chart	14
2.6	Conclusion	14
3	Methodology and Design	15
3.1	Section One	15
3.1.1	Subsection One	15
3.1.2	Subsection Two	16
3.1.2.1	Subsubsection One	16
3.1.2.2	Subsubsection Two	16
3.1.2.2.1	Paragraph a	16
3.1.2.2.2	Paragraph b	17
3.2	Section Two	17
3.2.1	Subsection One	17
3.2.2	Subsection Two	17
3.2.2.1	Subsubsection One	18
3.2.2.2	Subsubsection Two	18
3.2.2.2.1	Paragraph a	18
3.2.2.2.2	Paragraph b	18
4	Implementation and Results	20
4.1	Section One	20
4.1.1	Subsection One	20
4.1.2	Subsection Two	20
4.1.2.1	Subsubsection One	21
4.1.2.2	Subsubsection Two	21

4.1.2.2.1	Paragraph a	21
4.1.2.2.2	Paragraph b	21
4.2	Section Two	22
4.2.1	Subsection One	22
4.2.2	Subsection Two	22
4.2.2.1	Subsubsection One	22
4.2.2.2	Subsubsection Two	23
4.2.2.2.1	Paragraph a	23
4.2.2.2.2	Paragraph b	23
5	Evaluation and Discussion	24
5.1	Section One	24
5.1.1	Subsection One	24
5.1.2	Subsection Two	24
5.1.2.1	Subsubsection One	25
5.1.2.2	Subsubsection Two	25
5.1.2.2.1	Paragraph a	25
5.1.2.2.2	Paragraph b	25
5.2	Section Two	26
5.2.1	Subsection One	26
5.2.2	Subsection Two	26
5.2.2.1	Subsubsection One	26
5.2.2.2	Subsubsection Two	27
5.2.2.2.1	Paragraph a	27
5.2.2.2.2	Paragraph b	27
	General Conclusion and Perspectives	28
A	Glossary	29
B	Acronyms	30
C	Supplementary Material	31
	Bibliography	32

IMPORTANT CONFIDENTIALITY WARNING

This report is an Oracle internal document. It is strictly confidential. However, Soukaina Bouhissin and the examiners, as members of the ENSAM pedagogical team in charge of supervising this project, are authorized to consult it for the sole purpose of evaluating the author. Any use outside of this context is prohibited. If you have obtained a copy of this report without it being explicitly intended for you by the author, please delete it and report back to the author immediately. In addition, the examples described in this report are for illustrative purposes only and do not reflect Oracle's actual business activities.

General Context

1.1 Introduction

In today's world, businesses, technology companies, and the public sector rely on databases to store their data, while utilizing cloud services to manage their IT infrastructure and other essential tools such as ERP, CRM, SCM, EPM, and SPM. These technologies have taken a crucial role, making it inconceivable to function without them.

Oracle, a leading US multinational computer technology corporation, not only employs these tools in its day-to-day operations but also creates and provides them to customers. Moreover, the company places a high value on innovation and ensuring customer satisfaction.

1.2 Presentation of the Host Organization

1.2.1 Oracle Corporation

Oracle Corporation is a tech company with headquarters in Austin, Texas. In 1977, Larry Ellison, Bob Miner, and Ed Oates founded Oracle Corporation, then called Software Development Laboratories (SDL).

The company first focused on creating Oracle Database, a relational database management system (RDBMS), that became its flagship product. Enterprise software, cloud computing, hardware systems, and consulting services are now all part of Oracle's growing list of services and product options. Sun Microsystems was a key acquisition in 2010, giving Oracle access to technologies such as Java and the Solaris operating system.

Oracle is a leading provider of cloud-based IT infrastructure and software that helps organizations grow, find new sources of efficiency, and improve their performance. Oracle developed the first and only autonomous database in the world as well as Oracle Cloud.

1.2.2 Oracle Morocco R&D Center

The first Oracle R&D center in Africa and one of only a few globally, the Oracle Research & Development Center Morocco is a software development center established by Oracle Corporation in Casablanca, Morocco. The center was inaugurated in 2022 and is one of Oracle's key development centers in the EMEA region. The center is staffed by highly skilled and talented software developers, engineers, and project managers who work on a variety of projects



Figure 1.1: Oracle MADC Location

across various Oracle product lines utilizing the newest trends in innovation, such as artificial intelligence, augmented/virtual/mixed reality, big data analytics, Blockchain, cloud computing, cybersecurity, internet of things, machine learning, mobile computing, and serverless computing.

The center's researchers make use of all these technologies to tackle the most pressing problems facing business, science, and the public sector. Expansion of internship and graduate recruitment initiatives and joint research projects with regional universities are all part of this global program.

Oracle MADC has been successful in delivering several high-quality products and solutions, such as Oracle Cloud Infrastructure (OCI), Autonomous Data Warehouse (ADW), etc.



Figure 1.2: MADC Major Milestones

1.2.3 Business Area of Activity

Oracle provides a variety of goods and services to its clients. Software development tools, cloud services, training, consultancy, and credentials are just a few of the goods and services offered. One of the brands Oracle offers is Oracle Database, which has held the top spot since 1979 and is supported by cloud artificial intelligence. SaaS, PaaS, IaaS, and DaaS are just a few of the integrated solutions offered by Oracle Cloud Infrastructure for business, IT, and development needs. Oracle Middleware is a platform integrated for creating and running intelligent, agile applications while increasing technical efficiency using contemporary software designs. Oracle Services include Oracle Advanced Customer Support Services, Oracle Premium Support, Oracle Consulting, Oracle Financing, and Oracle Managed Cloud Services.

1.2.3.1 Oracle Morocco R&D Center Lines of Businesses (LOBs)

1.2.3.1.1 Oracle APEX The most widely used enterprise low-code application platform in the world, Oracle APEX lets you create safe, scalable, and feature-rich web and mobile apps that can be deployed on-premises or in the cloud. With APEX, developers can create and launch engaging applications that address practical issues and deliver benefits right away. It is not necessary to possess extensive knowledge of multiple technologies to provide complex solutions.

1.2.3.1.2 Oracle Health and Artificial Intelligence Oracle APEX helps transform the technologies that are revolutionizing healthcare. Oracle APEX solutions accelerate the development of the next generation of drugs, devices, and therapies through a cohesive ecosystem that harnesses data to deliver results. There are no more silos — just clinical trials, data, strategy, and insights flowing together for better outcomes.

1.2.3.1.3 Oracle Linux Oracle Linux delivers the operating system of choice for Oracle customers, Oracle Cloud, and Oracle Engineered Systems. Oracle has been an active contributor since 1998, shipping the first-ever commercial database on Linux along with multiple enhancements. Today, it is focused on developing open-source products while improving the stability and performance of the operating system.

1.2.3.1.4 Oracle NetSuite NetSuite is a cloud-based business management software that helps organizations manage their financials, operations, and customer relationships. It offers features such as accounting, CRM, inventory management, e-commerce, and more, all in one platform.

1.2.3.1.5 Oracle Database Development Tools The cloud and on-premises solutions drive the secure data access at a scale that powers the modern world. Combined with advanced data visualization and analysis, Oracle Database Development Tools enable data platforms for application development for any purpose. The products include Oracle SQL Developer, Oracle Data Modeler, Oracle REST Data Services, and Oracle SQLcl.

1.2.3.1.6 Fusion ERP Oracle Fusion Cloud ERP is a complete, modern, cloud ERP suite that provides teams with advanced capabilities, such as AI to automate manual processes, analytics to react to market shifts in real-time, and automatic updates to stay current and gain a competitive advantage.

1.2.3.1.7 Oracle Analytics Oracle Analytics is a comprehensive platform for diverse data workloads. It turns information into action with insights from cloud, on-premises, and hybrid data, empowering users to make business-defining decisions sooner through AI and ML-powered analytical applications.

1.2.3.1.8 Oracle Cloud Infrastructure Oracle Cloud is the first public cloud built from the ground up to be a better cloud for every application. By rethinking core engineering and systems design for cloud computing, it created innovations that solve problems customers have with existing public clouds and offers the complete services customers need to build innovative cloud applications.

1.2.3.1.9 Oracle Digital Government Group The data intelligence platform provides the insights that government leaders need to make the decisions that matter. The mission is to help national governments on their digital transformation journeys and improve citizens' experience with government services.

1.2.3.1.10 Oracle Java Platform Group The Oracle Java Platform Group is dedicated to advancing and maintaining the Java platform, a cornerstone of modern software development used by millions of developers worldwide. This group drives innovations in the Java ecosystem, ensuring the language remains robust, efficient, and versatile for a wide range of applications.

1.2.4 Organizational Structure

Oracle Corporation has a hierarchical organizational structure with three main business divisions: Hardware, Services, Cloud, and License. Each segment is divided into smaller business units focused on goods, services, or geographical areas. The Board of Directors oversees the company's performance and strategic direction, while the Executive Leadership

Team manages daily operations. Functional divisions, including sales, marketing, finance, human resources, legal, and IT, fall under the Executive Leadership Team.

Business units are arranged by region, product, and service, with a senior VP or general manager overseeing each unit's operations. The employees are subordinated to both a functional manager and a business unit manager in the matrix-based organizational structure of the business units.

In general, Oracle's organizational structure is built to encourage teamwork, creativity, and client attention. The matrix form enables people to work across many functions and business divisions to achieve shared goals, while the hierarchical structure promotes effective decision-making and communication.

1.2.5 Organizational Culture

Oracle has a strong corporate culture that prioritizes innovation, client focus, and excellence. The company's history of innovation, emphasis on customer satisfaction, and dedication to fostering a friendly and collaborative work environment all shape the company's culture.

Oracle's culture places a strong emphasis on the following values:

Innovation One of Oracle's defining characteristics. The business has a lengthy history of creating cutting-edge goods and technology that have revolutionized the tech sector, promoted by a dedication to R&D and by giving staff members the tools to test fresh concepts.

Customer Satisfaction Oracle's customer-centric approach reflects its commitment to providing goods and services that satisfy client needs. Employees are encouraged to collaborate closely with clients to comprehend their demands and create tailored solutions.

Quality Oracle is dedicated to providing quality operations, products, and services, reflected in its solid reputation for dependability and performance. The company expects every person to strive for excellence.

Collaboration and Teamwork Effective collaboration is essential for achieving Oracle's goals. The company encourages employees to work together across departments and functions, providing tools and resources to facilitate communication.

Overall, Oracle's organizational culture is characterized by a commitment to innovation, customer focus, excellence, collaboration, and teamwork. These values are reflected in the company's products, services, and operations, and are embraced by its employees at all levels of the organization.

1.3 Topic Context

1.3.1 General Context of the Project

Before diving into the core issue of the topic, it is useful to provide some background information. The team I worked with during this internship was called “Digital Government” — which directly reflects the primary objective this project aims to serve.

Digital government services — also referred to as e-government — refer to the use of information and communication technologies to improve the efficiency, effectiveness, and transparency of government operations. These systems reduce bureaucratic delays, minimise paperwork, and enhance the overall quality of public services.

E-government systems are commonly categorised based on the type of interaction they facilitate. Three main categories exist: Government-to-Government (G2G), Government-to-Citizen (G2C), and Government-to-Employee (G2E).

G2G refers to interactions between different government agencies or departments, whether at the same level or across levels of government. An example relevant to this project is the sharing of agricultural satellite data between a national space agency and a ministry of agriculture to inform food security decisions.

G2C focuses on delivering government services and information directly to citizens — making them more accessible, timely, and actionable. In the context of agriculture, this translates to providing smallholder farmers and local administrations with accurate, up-to-date crop production intelligence.

G2E refers to internal government process management and employee communication — for example, equipping field agents with data-driven tools to support their monitoring activities.

The Digital Government team at Oracle aims to leverage existing government data and modern technologies to solve real-world problems across sectors including agriculture, healthcare, and education. This project sits firmly within the agricultural domain and contributes to Oracle’s Government Data Intelligence for Agriculture platform — launched in September 2025 — which supports governments in building more resilient and data-driven food systems.

It is important to note that this project is part of an ongoing collaboration with different governments. The requirements and needs of each country differ, and Rwanda represents one of the early adopters of this approach.

1.3.2 Problematic

During my internship, my focus was on developing a machine learning pipeline to classify crop types across Rwanda using Sentinel-2 satellite time series data — with the broader aim of supporting agricultural monitoring and food security decision-making.

Agriculture plays a vital role in Rwanda's economy and food system. Accurate, timely knowledge of what crops are growing where — and how much — is essential for effective resource allocation, early warning of food shortfalls, and informed policy decisions. However, obtaining this information at national scale through traditional ground surveys is physically and financially impossible. Surveys are expensive, sparse, seasonal, and always outdated by the time results are available.

The challenge is compounded by the nature of smallholder agriculture in Rwanda. Farms are small, fragmented, and often intercropped — with multiple crops growing on the same plot within the same season. Satellite pixels at the boundary between fields capture mixed spectral signals from more than one crop, creating label ambiguity. Cloud cover — persistent throughout Rwanda's growing seasons — introduces temporal gaps in satellite observations that must be handled carefully. And the extreme class imbalance between dominant crops and minority crops like Sweet Potato — representing only 0.3% of the training data — makes reliable multi-class classification a significant methodological challenge.

1.3.3 Solution

To address these challenges, I designed and implemented an end-to-end multi-class crop classification pipeline using Sentinel-2 satellite time series data, processed on Oracle Cloud Infrastructure.

The pipeline begins with rigorous data preprocessing — including label noise correction, boundary pixel identification and removal, geometric quality filtering, cloud-based quality flagging, temporal gap analysis, and linear interpolation to reconstruct missing observations. Each step was grounded in peer-reviewed literature and validated against the data.

Eight vegetation indices were computed from the cleaned time series — capturing greenness, soil-adjusted vegetation, chlorophyll content, and moisture levels. From these, 50 summary features were extracted per pixel: 32 statistical features capturing the overall seasonal signal, and 18 phenological features capturing crop growth timing — start of season, end of season, peak, amplitude, rate of increase, and integral.

Feature selection was applied in three stages — ANOVA, Random Forest importance ranking, and Recursive Feature Elimination — reducing the feature set from 50 to 37 without loss of predictive performance. Class imbalance was addressed through a combination of balanced class weights, SMOTE oversampling of the minority class, and prediction threshold adjustment — all optimised through grid search.

Two models were trained and compared: Random Forest as a baseline and XGBoost as the primary model. XGBoost outperformed Random Forest across all evaluation metrics — Macro F1, Weighted F1, G-mean, and Cohen's Kappa — with particularly significant gains on Sweet Potato classification, where F1 improved from 0.35 to 0.64.

This approach demonstrated that reliable multi-class crop mapping is achievable from free

satellite data at national scale — with no field surveys required — and provides a foundation for integration into Oracle’s Government Data Intelligence for Agriculture platform.

1.4 Conclusion

This chapter introduced the general context of the internship project. It first presented Oracle as the host organization, including its activities, structure, culture, and the role of the Oracle Morocco Research and Development Center. It then described the Digital Government context in which the project was conducted, highlighting the importance of e-government services and their role in improving public-sector decision-making.

The chapter also presented the main problem addressed during the internship: the difficulty of producing accurate and timely crop-type information at national scale using traditional field-survey methods. Finally, it introduced the proposed satellite-based machine learning solution, which aims to support agricultural monitoring and food security decision-making in Rwanda.

Project Management

2.1 Introduction

This chapter presents the organizational aspects of the project carried out during my internship. It describes the methods, tools, and practices used to structure the work, improve collaboration, and ensure efficient task completion.

Effective project organization is an essential factor in the success of any technical project. It allows teams to collaborate smoothly, communicate clearly, and deliver high-quality results within the expected timeframe. Throughout this chapter, the internship process, the project management tools, the adopted methodology, and the project planning are presented.

2.2 Internship Process

This section describes the main phases of my internship at Oracle, from the onboarding period to the technical training.

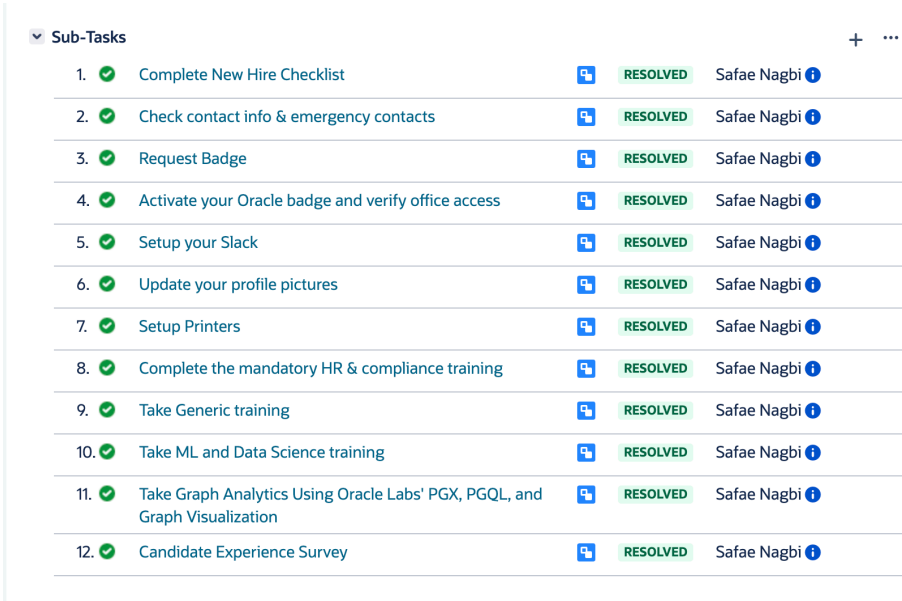
2.2.1 Onboarding Phase

The initial phase of my internship at Oracle consisted of becoming familiar with the company's internal environment and the Morocco Research and Development Center (MADC). This phase included meetings with key people, such as the HR manager, the program manager, members of the Oracle Morocco team, my mentor, and the project team.

During this period, I also received my internship equipment and participated in introductory sessions that helped me understand the objectives of the project, the team organization, and the expectations related to my work. After receiving my workstation, I configured the required working environment, including SSO, VPN, and BitLocker.

2.2.2 Training Phase

I completed a comprehensive compliance training program covering essential company policies, security rules, and internal regulations. This training provided a solid foundation for the internship and helped ensure that I was aware of my responsibilities and Oracle's expectations.



Sub-Tasks				
1.	✓	Complete New Hire Checklist	RESOLVED	Safae Nagbi
2.	✓	Check contact info & emergency contacts	RESOLVED	Safae Nagbi
3.	✓	Request Badge	RESOLVED	Safae Nagbi
4.	✓	Activate your Oracle badge and verify office access	RESOLVED	Safae Nagbi
5.	✓	Setup your Slack	RESOLVED	Safae Nagbi
6.	✓	Update your profile pictures	RESOLVED	Safae Nagbi
7.	✓	Setup Printers	RESOLVED	Safae Nagbi
8.	✓	Complete the mandatory HR & compliance training	RESOLVED	Safae Nagbi
9.	✓	Take Generic training	RESOLVED	Safae Nagbi
10.	✓	Take ML and Data Science training	RESOLVED	Safae Nagbi
11.	✓	Take Graph Analytics Using Oracle Labs' PGX, PGQL, and Graph Visualization	RESOLVED	Safae Nagbi
12.	✓	Candidate Experience Survey	RESOLVED	Safae Nagbi

Figure 2.1: Jira Ticket Example

Following this compliance training, I started a series of generic technical training sessions focused on strengthening my foundations in several areas, including Oracle Cloud Infrastructure (OCI), DevOps, and Linux.

Placeholder for Generic Training figure

Figure 2.2: Generic Training

Once the foundational training sessions were completed, I moved on to more specific training related to the project requirements. This included APEX foundations and the development of small applications in order to practice and better understand the tools used within the team.

2.3 Management Tools

To facilitate collaboration, communication, documentation, and task tracking, several management tools were used throughout the internship. The main tools are presented in the following sections.

2.3.1 Confluence

Confluence provided a centralized platform for knowledge sharing and documentation. It was used to document tasks, research, technical notes, and project-related information throughout the internship. It also served as a repository where useful project details and references could be stored and accessed by the team [1].

2.3.2 Zoom

Zoom was used as a video conferencing platform to participate in virtual meetings with the team. It offered a practical and efficient solution for remote communication and collaboration, allowing clear discussions and regular follow-up meetings [2].

2.3.3 Jira

Jira, a project management and issue-tracking tool developed by Atlassian, was used to organize tasks, track progress, assign work, set priorities, and manage project workflows. It allowed the team to follow the status of different tasks and ensure that deadlines were respected.

Jira supports several project management methodologies, including Agile, Scrum, and Kanban. It provides features such as boards, backlogs, issues, and sprints, which helped structure the work and monitor progress throughout the internship [3].

2.3.4 Slack

Slack was used as a centralized communication hub for the team. It enabled real-time information sharing, discussions about project updates, direct messaging, and file sharing. Its channel-based organization helped keep conversations structured and accessible.

In addition, the search functionality in Slack channels was particularly useful for debugging and problem solving. It made it possible to quickly retrieve previous discussions or similar internal errors shared by other team members, which saved significant time during development [4].

2.4 Project Methodology

Throughout the internship, the team followed the Scrum framework, which is one of the most widely used frameworks within Agile methodology. Scrum helped structure the project into iterative development cycles, allowing the team to adapt to changing requirements while maintaining a clear focus on delivering valuable increments.

To illustrate this methodology, consider the example of building a simple to-do list application. If the product owner prioritizes two features for the first sprint, such as adding new tasks and marking tasks as completed, the team focuses on implementing these features during the sprint. During development, team members review each other's work and test the features to verify that they meet the acceptance criteria defined at the beginning. At the end of the sprint, a functional increment of the application is delivered.

In this project, the product backlog was prioritized according to the value and relevance of the tasks. Since some requirements were still evolving and could change over time, Scrum

was an appropriate approach. It supported flexibility, continuous feedback, and progressive improvement.

Furthermore, the Scrum framework encouraged communication, collaboration, and continuous improvement. By following Agile principles, the team was able to respond to change, iterate quickly, and work toward a high-quality solution aligned with stakeholder expectations.

The main Agile principles followed during the project are:

1. Customer satisfaction through early and continuous delivery of valuable software.
2. Welcoming changing requirements, even late in development, in order to support the customer's competitive advantage.
3. Delivering working software frequently, with a preference for shorter timescales.
4. Maintaining collaboration between business stakeholders and developers throughout the project.
5. Building projects around motivated individuals by giving them the environment and support they need and trusting them to get the job done.
6. Prioritizing face-to-face conversation as an efficient and effective method of communication within the development team.
7. Considering working software as the primary measure of progress.
8. Promoting sustainable development so that sponsors, developers, and users can maintain a constant pace.
9. Maintaining continuous attention to technical excellence and good design.
10. Valuing simplicity, which means maximizing the amount of work not done.
11. Allowing the best architectures, requirements, and designs to emerge from self-organizing teams.
12. Reflecting regularly on how to become more effective, then adjusting the team's behavior accordingly [5].

In my case, I also had additional meetings with different team members. These meetings were mainly opportunities to discuss tasks in more detail, clarify technical points, and follow the progress of the work.

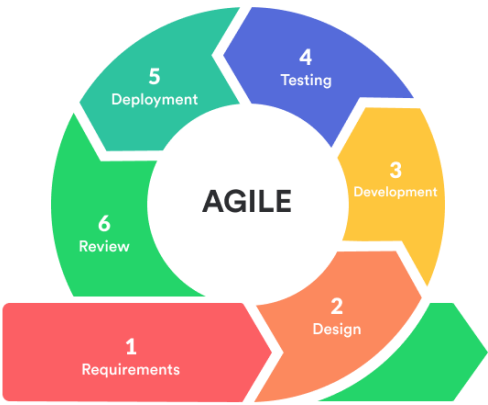


Figure 2.3: Agile Methodology

2.5 Gantt Chart

A Gantt chart, named after its creator Henry L. Gantt, is a visual tool used in project management to represent tasks along a timeline. It uses horizontal bars to show the duration of each task, the dependencies between tasks, and the overall schedule of a project.

This tool makes complex project planning easier to understand by presenting tasks, milestones, and deadlines in a clear visual format. It helps project managers and team members monitor progress, identify delays, and keep the project aligned with its expected timeline.

In the context of this internship, the Gantt chart summarizes the main phases of the project during the first four months, from the onboarding process to the writing of this report.

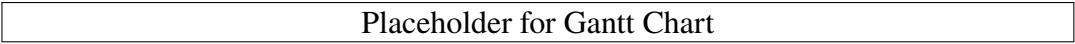


Figure 2.4: Gantt Chart

2.6 Conclusion

This chapter presented the organizational dimensions of the project. It described the internship process, including onboarding and training, as well as the tools used to support communication, documentation, and task management.

The chapter also introduced the Scrum methodology adopted during the project and explained how Agile principles helped structure the work and support collaboration. Finally, the Gantt chart provided an overview of the project timeline. These organizational elements contributed to a clearer workflow and helped ensure the successful progress of the internship project.

Methodology and Design

Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Aenean nonummy turpis id odio. Integer euismod imperdiet turpis. Ut nec leo nec diam imperdiet lacinia. Etiam eget lacus eget mi ultricies posuere. In placerat tristique tortor. Sed porta vestibulum metus. Nulla iaculis sollicitudin pede. Fusce luctus tellus in dolor. Curabitur auctor velit a sem. Morbi sapien. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec adipiscing urna vehicula nunc. Sed ornare leo in leo. In rhoncus leo ut dui. Aenean dolor quam, volutpat nec, fringilla id, consectetur vel, pede.

3.1 Section One

Nulla malesuada risus ut urna. Aenean pretium velit sit amet metus. Duis iaculis. In hac habitasse platea dictumst. Nullam molestie turpis eget nisl. Duis a massa id pede dapibus ultricies. Sed eu leo. In at mauris sit amet tortor bibendum varius. Phasellus justo risus, posuere in, sagittis ac, varius vel, tortor. Quisque id enim. Phasellus consequat, libero pretium nonummy fringilla, tortor lacus vestibulum nunc, ut rhoncus ligula neque id justo. Nullam accumsan euismod nunc. Proin vitae ipsum ac metus dictum tempus. Nam ut wisi. Quisque tortor felis, interdum ac, sodales a, semper a, sem. Curabitur in velit sit amet dui tristique sodales. Vivamus mauris pede, lacinia eget, pellentesque quis, scelerisque eu, est. Aliquam risus. Quisque bibendum pede eu dolor.

3.1.1 Subsection One

Donec tempus neque vitae est. Aenean egestas odio sed risus ullamcorper ullamcorper. Sed in nulla a tortor tincidunt egestas. Nam sapien tortor, elementum sit amet, aliquam in, porttitor faucibus, enim. Nullam congue suscipit nibh. Quisque convallis. Praesent arcu nibh, vehicula eget, accumsan eu, tincidunt a, nibh. Suspendisse vulputate, tortor quis adipiscing viverra, lacus nibh dignissim tellus, eu suscipit risus ante fringilla diam. Quisque a libero vel pede imperdiet aliquet. Pellentesque nunc nibh, eleifend a, consequat consequat, hendrerit nec, diam. Sed urna. Maecenas laoreet eleifend neque. Vivamus purus odio, eleifend non, iaculis a, ultrices sit amet, urna. Mauris faucibus odio vitae risus. In nisl. Praesent purus. Integer

iaculis, sem eu egestas lacinia, lacus pede scelerisque augue, in ullamcorper dolor eros ac lacus. Nunc in libero.

3.1.2 Subsection Two

Fusce suscipit cursus sem. Vivamus risus mi, egestas ac, imperdiet varius, faucibus quis, leo. Aenean tincidunt. Donec suscipit. Cras id justo quis nibh scelerisque dignissim. Aliquam sagittis elementum dolor. Aenean consectetur justo in pede. Curabitur ullamcorper ligula nec orci. Aliquam purus turpis, aliquam id, ornare vitae, porttitor non, wisi. Maecenas luctus porta lorem. Donec vitae ligula eu ante pretium varius. Proin tortor metus, convallis et, hendrerit non, scelerisque in, urna. Cras quis libero eu ligula bibendum tempor. Vivamus tellus quam, malesuada eu, tempus sed, tempor sed, velit. Donec lacinia auctor libero.

3.1.2.1 Subsubsection One

Praesent sed neque id pede mollis rutrum. Vestibulum iaculis risus. Pellentesque lacus. Ut quis nunc sed odio malesuada egestas. Duis a magna sit amet ligula tristique pretium. Ut pharetra. Vestibulum imperdiet magna nec wisi. Mauris convallis. Sed accumsan sollicitudin massa. Sed id enim. Nunc pede enim, lacinia ut, pulvinar quis, suscipit semper, elit. Cras accumsan erat vitae enim. Cras sollicitudin. Vestibulum rutrum blandit massa.

3.1.2.2 Subsubsection Two

Sed gravida lectus ut purus. Morbi laoreet magna. Pellentesque eu wisi. Proin turpis. Integer sollicitudin augue nec dui. Fusce lectus. Vivamus faucibus nulla nec lacus. Integer diam. Pellentesque sodales, enim feugiat cursus volutpat, sem mauris dignissim mauris, quis consequat sem est fermentum ligula. Nullam justo lectus, condimentum sit amet, posuere a, fringilla mollis, felis. Morbi nulla nibh, pellentesque at, nonummy eu, sollicitudin nec, ipsum. Cras neque. Nunc augue. Nullam vitae quam id quam pulvinar blandit. Nunc sit amet orci. Aliquam erat elit, pharetra nec, aliquet a, gravida in, mi. Quisque urna enim, viverra quis, suscipit quis, tincidunt ut, sapien. Cras placerat consequat sem. Curabitur ac diam. Curabitur diam tortor, mollis et, viverra ac, tempus vel, metus.

3.1.2.2.1 Paragraph a Curabitur ac lorem. Vivamus non justo in dui mattis posuere. Etiam accumsan ligula id pede. Maecenas tincidunt diam nec velit. Praesent convallis sapien ac est. Aliquam ullamcorper euismod nulla. Integer mollis enim vel tortor. Nulla sodales placerat nunc. Sed tempus rutrum wisi. Duis accumsan gravida purus. Nunc nunc. Etiam facilisis dui eu sem. Vestibulum semper. Praesent eu eros. Vestibulum tellus nisl, dapibus id, vestibulum sit amet, placerat ac, mauris. Maecenas et elit ut erat placerat dictum. Nam

feugiat, turpis et sodales volutpat, wisi quam rhoncus neque, vitae aliquam ipsum sapien vel enim. Maecenas suscipit cursus mi.

3.1.2.2.2 Paragraph b Quisque consectetur. In suscipit mauris a dolor pellentesque consectetur. Mauris convallis neque non erat. In lacinia. Pellentesque leo eros, sagittis quis, fermentum quis, tincidunt ut, sapien. Maecenas sem. Curabitur eros odio, interdum eu, feugiat eu, porta ac, nisl. Curabitur nunc. Etiam fermentum convallis velit. Pellentesque laoreet lacus. Quisque sed elit. Nam quis tellus. Aliquam tellus arcu, adipiscing non, tincidunt eleifend, adipiscing quis, augue. Vivamus elementum placerat enim. Suspendisse ut tortor. Integer faucibus adipiscing felis. Aenean consectetur mattis lectus. Morbi malesuada faucibus dolor. Nam lacus. Etiam arcu libero, malesuada vitae, aliquam vitae, blandit tristique, nisl.

3.2 Section Two

Maecenas accumsan dapibus sapien. Duis pretium iaculis arcu. Curabitur ut lacus. Aliquam vulputate. Suspendisse ut purus sed sem tempor rhoncus. Ut quam dui, fringilla at, dictum eget, ultricies quis, quam. Etiam sem est, pharetra non, vulputate in, pretium at, ipsum. Nunc semper sagittis orci. Sed scelerisque suscipit diam. Ut volutpat, dolor at ullamcorper tristique, eros purus mollis quam, sit amet ornare ante nunc et enim.

3.2.1 Subsection One

Phasellus fringilla, metus id feugiat consectetur, lacus wisi ultrices tellus, quis lobortis nibh lorem quis tortor. Donec egestas ornare nulla. Mauris mi tellus, porta faucibus, dictum vel, nonummy in, est. Aliquam erat volutpat. In tellus magna, porttitor lacinia, molestie vitae, pellentesque eu, justo. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Sed orci nibh, scelerisque sit amet, suscipit sed, placerat vel, diam. Vestibulum nonummy vulputate orci. Donec et velit ac arcu interdum semper. Morbi pede orci, cursus ac, elementum non, vehicula ut, lacus. Cras volutpat. Nam vel wisi quis libero venenatis placerat. Aenean sed odio. Quisque posuere purus ac orci. Vivamus odio. Vivamus varius, nulla sit amet semper viverra, odio mauris consequat lacus, at vestibulum neque arcu eu tortor. Donec iaculis tincidunt tellus. Aliquam erat volutpat. Curabitur magna lorem, dignissim volutpat, viverra et, adipiscing nec, dolor. Praesent lacus mauris, dapibus vitae, sollicitudin sit amet, nonummy eget, ligula.

3.2.2 Subsection Two

Cras egestas ipsum a nisl. Vivamus varius dolor ut dolor. Fusce vel enim. Pellentesque accumsan ligula et eros. Cras id lacus non tortor facilisis facilisis. Etiam nisl elit, cursus

sed, fringilla in, congue nec, urna. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer at turpis. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Duis fringilla, ligula sed porta fringilla, ligula wisi commodo felis, ut adipiscing felis dui in enim. Suspendisse malesuada ultrices ante. Pellentesque scelerisque augue sit amet urna. Nulla volutpat aliquet tortor. Cras aliquam, tellus at aliquet pellentesque, justo sapien commodo leo, id rhoncus sapien quam at erat. Nulla commodo, wisi eget sollicitudin pretium, orci orci aliquam orci, ut cursus turpis justo et lacus. Nulla vel tortor. Quisque erat elit, viverra sit amet, sagittis eget, porta sit amet, lacus.

3.2.2.1 Subsubsection One

In hac habitasse platea dictumst. Proin at est. Curabitur tempus vulputate elit. Pellentesque sem. Praesent eu sapien. Duis elit magna, aliquet at, tempus sed, vehicula non, enim. Morbi viverra arcu nec purus. Vivamus fringilla, enim et commodo malesuada, tortor metus elementum ligula, nec aliquet est sapien ut lectus. Aliquam mi. Ut nec elit. Fusce euismod luctus tellus. Curabitur scelerisque. Nullam purus. Nam ultricies accumsan magna. Morbi pulvinar lorem sit amet ipsum. Donec ut justo vitae nibh mollis congue. Fusce quis diam. Praesent tempus eros ut quam.

3.2.2.2 Subsubsection Two

Donec in nisl. Fusce vitae est. Vivamus ante ante, mattis laoreet, posuere eget, congue vel, nunc. Fusce sem. Nam vel orci eu eros viverra luctus. Pellentesque sit amet augue. Nunc sit amet ipsum et lacus varius nonummy. Integer rutrum sem eget wisi. Aenean eu sapien. Quisque ornare dignissim mi. Duis a urna vel risus pharetra imperdiet. Suspendisse potenti.

3.2.2.2.1 Paragraph a Morbi justo. Aenean nec dolor. In hac habitasse platea dictumst. Proin nonummy porttitor velit. Sed sit amet leo nec metus rhoncus varius. Cras ante. Vestibulum commodo sem tincidunt massa. Nam justo. Aenean luctus, felis et condimentum lacinia, lectus enim pulvinar purus, non porta velit nisl sed eros. Suspendisse consequat. Mauris a dui et tortor mattis pretium. Sed nulla metus, volutpat id, aliquam eget, ullamcorper ut, ipsum. Morbi eu nunc. Praesent pretium. Duis aliquam pulvinar ligula. Ut blandit egestas justo. Quisque posuere metus viverra pede.

3.2.2.2.2 Paragraph b Vivamus sodales elementum neque. Vivamus dignissim accumsan neque. Sed at enim. Vestibulum nonummy interdum purus. Mauris ornare velit id nibh pretium ultricies. Fusce tempor pellentesque odio. Vivamus augue purus, laoreet in, scelerisque vel, commodo id, wisi. Duis enim. Nulla interdum, nunc eu semper eleifend, enim dolor pretium elit, ut commodo ligula nisl a est. Vivamus ante. Nulla leo massa, posuere nec, volutpat vitae, rhoncus eu, magna.

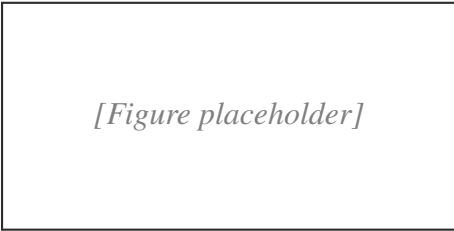


Figure 3.1: Sample figure in Chapter 3

Table 3.1: Sample table in Chapter 3

Column A	Column B	Column C
Item 1	Description	42
Item 2	Description	87

```
1 # Placeholder code
2 def hello_world():
3     """Print a greeting."""
4     print("Hello, World!")
5
6 hello_world()
```

Listing 3.1: Sample Python listing

Implementation and Results

Quisque facilisis auctor sapien. Pellentesque gravida hendrerit lectus. Mauris rutrum sodales sapien. Fusce hendrerit sem vel lorem. Integer pellentesque massa vel augue. Integer elit tortor, feugiat quis, sagittis et, ornare non, lacus. Vestibulum posuere pellentesque eros. Quisque venenatis ipsum dictum nulla. Aliquam quis quam non metus eleifend interdum. Nam eget sapien ac mauris malesuada adipiscing. Etiam eleifend neque sed quam. Nulla facilisi. Proin a ligula. Sed id dui eu nibh egestas tincidunt. Suspendisse arcu.

4.1 Section One

Maecenas dui. Aliquam volutpat auctor lorem. Cras placerat est vitae lectus. Curabitur massa lectus, rutrum euismod, dignissim ut, dapibus a, odio. Ut eros erat, vulputate ut, interdum non, porta eu, erat. Cras fermentum, felis in porta congue, velit leo facilisis odio, vitae consectetur lorem quam vitae orci. Sed ultrices, pede eu placerat auctor, ante ligula rutrum tellus, vel posuere nibh lacus nec nibh. Maecenas laoreet dolor at enim. Donec molestie dolor nec metus. Vestibulum libero. Sed quis erat. Sed tristique. Duis pede leo, fermentum quis, consectetur eget, vulputate sit amet, erat.

4.1.1 Subsection One

Donec vitae velit. Suspendisse porta fermentum mauris. Ut vel nunc non mauris pharetra varius. Duis consequat libero quis urna. Maecenas at ante. Vivamus varius, wisi sed egestas tristique, odio wisi luctus nulla, lobortis dictum dolor ligula in lacus. Vivamus aliquam, urna sed interdum porttitor, metus orci interdum odio, sit amet euismod lectus felis et leo. Praesent ac wisi. Nam suscipit vestibulum sem. Praesent eu ipsum vitae pede cursus venenatis. Duis sed odio. Vestibulum eleifend. Nulla ut massa. Proin rutrum mattis sapien. Curabitur dictum gravida ante.

4.1.2 Subsection Two

Phasellus placerat vulputate quam. Maecenas at tellus. Pellentesque neque diam, dignissim ac, venenatis vitae, consequat ut, lacus. Nam nibh. Vestibulum fringilla arcu mollis arcu. Sed et turpis. Donec sem tellus, volutpat et, varius eu, commodo sed, lectus. Lorem ipsum dolor

sit amet, consectetur adipiscing elit. Quisque enim arcu, suscipit nec, tempus at, imperdiet vel, metus. Morbi volutpat purus at erat. Donec dignissim, sem id semper tempus, nibh massa eleifend turpis, sed pellentesque wisi purus sed libero. Nullam lobortis tortor vel risus. Pellentesque consequat nulla eu tellus. Donec velit. Aliquam fermentum, wisi ac rhoncus iaculis, tellus nunc malesuada orci, quis volutpat dui magna id mi. Nunc vel ante. Duis vitae lacus. Cras nec ipsum.

4.1.2.1 Subsubsection One

Morbi nunc. Aliquam consectetur varius nulla. Phasellus eros. Cras dapibus porttitor risus. Maecenas ultrices mi sed diam. Praesent gravida velit at elit vehicula porttitor. Phasellus nisl mi, sagittis ac, pulvinar id, gravida sit amet, erat. Vestibulum est. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur id sem elementum leo rutrum hendrerit. Ut at mi. Donec tincidunt faucibus massa. Sed turpis quam, sollicitudin a, hendrerit eget, pretium ut, nisl. Duis hendrerit ligula. Nunc pulvinar congue urna.

4.1.2.2 Subsubsection Two

Nunc velit. Nullam elit sapien, eleifend eu, commodo nec, semper sit amet, elit. Nulla lectus risus, condimentum ut, laoreet eget, viverra nec, odio. Proin lobortis. Curabitur dictum arcu vel wisi. Cras id nulla venenatis tortor congue ultrices. Pellentesque eget pede. Sed eleifend sagittis elit. Nam sed tellus sit amet lectus ullamcorper tristique. Mauris enim sem, tristique eu, accumsan at, scelerisque vulputate, neque. Quisque lacus. Donec et ipsum sit amet elit nonummy aliquet. Sed viverra nisl at sem. Nam diam. Mauris ut dolor. Curabitur ornare tortor cursus velit.

4.1.2.2.1 Paragraph a Morbi tincidunt posuere arcu. Cras venenatis est vitae dolor. Vivamus scelerisque semper mi. Donec ipsum arcu, consequat scelerisque, viverra id, dictum at, metus. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut pede sem, tempus ut, porttitor bibendum, molestie eu, elit. Suspendisse potenti. Sed id lectus sit amet purus faucibus vehicula. Praesent sed sem non dui pharetra interdum. Nam viverra ultrices magna.

4.1.2.2.2 Paragraph b Aenean laoreet aliquam orci. Nunc interdum elementum urna. Quisque erat. Nullam tempor neque. Maecenas velit nibh, scelerisque a, consequat ut, viverra in, enim. Duis magna. Donec odio neque, tristique et, tincidunt eu, rhoncus ac, nunc. Mauris malesuada malesuada elit. Etiam lacus mauris, pretium vel, blandit in, ultricies id, libero. Phasellus bibendum erat ut diam. In congue imperdiet lectus.

4.2 Section Two

Aenean scelerisque. Fusce pretium porttitor lorem. In hac habitasse platea dictumst. Nulla sit amet nisl at sapien egestas pretium. Nunc non tellus. Vivamus aliquet. Nam adipiscing euismod dolor. Aliquam erat volutpat. Nulla ut ipsum. Quisque tincidunt auctor augue. Nunc imperdiet ipsum eget elit. Aliquam quam leo, consectetur non, ornare sit amet, tristique quis, felis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Pellentesque interdum quam sit amet mi. Pellentesque mauris dui, dictum a, adipiscing ac, fermentum sit amet, lorem.

4.2.1 Subsection One

Ut quis wisi. Praesent quis massa. Vivamus egestas risus eget lacus. Nunc tincidunt, risus quis bibendum facilisis, lorem purus rutrum neque, nec porta tortor urna quis orci. Aenean aliquet, libero semper volutpat luctus, pede erat lacinia augue, quis rutrum sem ipsum sit amet pede. Vestibulum aliquet, nibh sed iaculis sagittis, odio dolor blandit augue, eget mollis urna tellus id tellus. Aenean aliquet aliquam nunc. Nulla ultricies justo eget orci. Phasellus tristique fermentum leo. Sed massa metus, sagittis ut, semper ut, pharetra vel, erat. Aliquam quam turpis, egestas vel, elementum in, egestas sit amet, lorem. Duis convallis, wisi sit amet mollis molestie, libero mauris porta dui, vitae aliquam arcu turpis ac sem. Aliquam aliquet dapibus metus.

4.2.2 Subsection Two

Vivamus commodo eros eleifend dui. Vestibulum in leo eu erat tristique mattis. Cras at elit. Cras pellentesque. Nullam id lacus sit amet libero aliquet hendrerit. Proin placerat, mi non elementum laoreet, eros elit tincidunt magna, a rhoncus sem arcu id odio. Nulla eget leo a leo egestas facilisis. Curabitur quis velit. Phasellus aliquam, tortor nec ornare rhoncus, purus urna posuere velit, et commodo risus tellus quis tellus. Vivamus leo turpis, tempus sit amet, tristique vitae, laoreet quis, odio. Proin scelerisque bibendum ipsum. Etiam nisl. Praesent vel dolor. Pellentesque vel magna. Curabitur urna. Vivamus congue urna in velit. Etiam ullamcorper elementum dui. Praesent non urna. Sed placerat quam non mi. Pellentesque diam magna, ultricies eget, ultrices placerat, adipiscing rutrum, sem.

4.2.2.1 Subsubsection One

Morbi sem. Nulla facilisi. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Nulla facilisi. Morbi sagittis ultrices libero. Praesent eu ligula sed sapien auctor sagittis. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Donec vel nunc. Nunc fermentum, lacus id aliquam porta, dui tortor

euismod eros, vel molestie ipsum purus eu lacus. Vivamus pede arcu, euismod ac, tempus id, pretium et, lacus. Curabitur sodales dapibus urna. Nunc eu sapien. Donec eget nunc a pede dictum pretium. Proin mauris. Vivamus luctus libero vel nibh.

4.2.2.2 Subsubsection Two

Fusce tristique risus id wisi. Integer molestie massa id sem. Vestibulum vel dolor. Pellentesque vel urna vel risus ultricies elementum. Quisque sapien urna, blandit nec, iaculis ac, viverra in, odio. In hac habitasse platea dictumst. Morbi neque lacus, convallis vitae, commodo ac, fermentum eu, velit. Sed in orci. In fringilla turpis non arcu. Donec in ante. Phasellus tempor feugiat velit. Aenean varius massa non turpis. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae;

4.2.2.2.1 Paragraph a Aliquam tortor. Morbi ipsum massa, imperdiet non, consectetur vel, feugiat vel, lorem. Quisque eget lorem nec elit malesuada vestibulum. Quisque sollicitudin ipsum vel sem. Nulla enim. Proin nonummy felis vitae felis. Nullam pellentesque. Duis rutrum feugiat felis. Mauris vel pede sed libero tincidunt mollis. Phasellus sed urna rhoncus diam euismod bibendum. Phasellus sed nisl. Integer condimentum justo id orci iaculis varius. Quisque et lacus. Phasellus elementum, justo at dignissim auctor, wisi odio lobortis arcu, sed sollicitudin felis felis eu neque. Praesent at lacus.

4.2.2.2.2 Paragraph b Vivamus sit amet pede. Duis interdum, nunc eget rutrum dignissim, nisl diam luctus leo, et tincidunt velit nisl id tellus. In lorem tellus, aliquet vitae, porta in, aliquet sed, lectus. Phasellus sodales. Ut varius scelerisque erat. In vel nibh eu eros imperdiet rutrum. Donec ac odio nec neque vulputate suscipit. Nam nec magna. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nullam porta, odio et sagittis iaculis, wisi neque fringilla sapien, vel commodo lorem lorem id elit. Ut sem lectus, scelerisque eget, placerat et, tincidunt scelerisque, ligula. Pellentesque non orci.

Evaluation and Discussion

Etiam vel ipsum. Morbi facilisis vestibulum nisl. Praesent cursus laoreet felis. Integer adipiscing pretium orci. Nulla facilisi. Quisque posuere bibendum purus. Nulla quam mauris, cursus eget, convallis ac, molestie non, enim. Aliquam congue. Quisque sagittis nonummy sapien. Proin molestie sem vitae urna. Maecenas lorem. Vivamus viverra consequat enim.

5.1 Section One

Nunc sed pede. Praesent vitae lectus. Praesent neque justo, vehicula eget, interdum id, facilisis et, nibh. Phasellus at purus et libero lacinia dictum. Fusce aliquet. Nulla eu ante placerat leo semper dictum. Mauris metus. Curabitur lobortis. Curabitur sollicitudin hendrerit nunc. Donec ultrices lacus id ipsum.

5.1.1 Subsection One

Donec a nibh ut elit vestibulum tristique. Integer at pede. Cras volutpat varius magna. Phasellus eu wisi. Praesent risus justo, lobortis eget, scelerisque ac, aliquet in, dolor. Proin id leo. Nunc iaculis, mi vitae accumsan commodo, neque sem lacinia nulla, quis vestibulum justo sem in eros. Quisque sed massa. Morbi lectus ipsum, vulputate a, mollis ut, accumsan placerat, tellus. Nullam in wisi. Vivamus eu ligula a nunc accumsan congue. Suspendisse ac libero. Aliquam erat volutpat. Donec augue. Nunc venenatis fringilla nibh. Fusce accumsan pulvinar justo. Nullam semper, dui ut dignissim auctor, orci libero fringilla massa, blandit pulvinar pede tortor id magna. Nunc adipiscing justo sed velit tincidunt fermentum.

5.1.2 Subsection Two

Integer placerat. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Sed in massa. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Phasellus tempus aliquam risus. Aliquam rutrum purus at metus. Donec posuere odio at erat. Nam non nibh. Phasellus ligula. Quisque venenatis lectus in augue. Sed vestibulum dapibus neque.

5.1.2.1 Subsubsection One

Mauris tempus eros at nulla. Sed quis dui dignissim mauris pretium tincidunt. Mauris ac purus. Phasellus ac libero. Etiam dapibus iaculis nunc. In lectus wisi, elementum eu, sollicitudin nec, imperdiet quis, dui. Nulla viverra neque ac libero. Mauris urna leo, adipiscing eu, ultrices non, blandit eu, dui. Maecenas dui neque, suscipit sit amet, rutrum a, laoreet in, eros. Ut eu nibh. Fusce nec erat tempus urna fringilla tempus. Curabitur id enim. Sed ante. Cras sodales enim sit amet wisi. Nunc fermentum consequat quam.

5.1.2.2 Subsubsection Two

Ut auctor, augue porta dignissim vestibulum, arcu diam lobortis velit, vel scelerisque risus augue sagittis risus. Maecenas eu justo. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris congue ligula eget tortor. Nullam laoreet urna sed enim. Donec eget eros ut eros volutpat convallis. Praesent turpis. Integer mauris diam, elementum quis, egestas ac, rutrum vel, orci. Nulla facilisi. Quisque adipiscing, nulla vitae elementum porta, sem urna volutpat leo, sed porta enim risus sed massa. Integer ac enim quis diam sodales luctus. Ut eget eros a ligula commodo ultricies. Donec eu urna viverra dolor hendrerit feugiat. Aliquam ac orci vel eros congue pharetra. Quisque rhoncus, justo eu volutpat faucibus, augue leo posuere lacus, a rhoncus purus pede vel est. Proin ultrices enim.

5.1.2.2.1 Paragraph a Aenean tincidunt laoreet dui. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Integer ipsum lectus, fermentum ac, malesuada in, eleifend ut, lorem. Vivamus ipsum turpis, elementum vel, hendrerit ut, semper at, metus. Vivamus sapien tortor, eleifend id, dapibus in, egestas et, pede. Pellentesque faucibus. Praesent lorem neque, dignissim in, facilisis nec, hendrerit vel, odio. Nam at diam ac neque aliquet viverra. Morbi dapibus ligula sagittis magna. In lobortis. Donec aliquet ultricies libero. Nunc dictum vulputate purus. Morbi varius. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In tempor. Phasellus commodo porttitor magna. Curabitur vehicula odio vel dolor.

5.1.2.2.2 Paragraph b Praesent facilisis, augue a adipiscing venenatis, libero risus molestie odio, pulvinar consectetur felis erat ac mauris. Nam vestibulum rhoncus quam. Sed velit urna, pharetra eu, eleifend eu, viverra at, wisi. Maecenas ultrices nibh at turpis. Aenean quam. Nulla ipsum. Aliquam posuere luctus erat. Curabitur magna felis, lacinia et, tristique id, ultrices ut, mauris. Suspendisse feugiat. Cras eleifend wisi vitae tortor. Phasellus leo purus, mattis sit amet, auctor in, rutrum in, magna. In hac habitasse platea dictumst. Phasellus imperdiet metus in sem. Vestibulum ac enim non sem ultricies sagittis. Sed vel diam.

5.2 Section Two

Integer vel enim sed turpis adipiscing bibendum. Vestibulum pede dolor, laoreet nec, posuere in, nonummy in, sem. Donec imperdiet sapien placerat erat. Donec viverra. Aliquam eros. Nunc consequat massa id leo. Sed ullamcorper, lorem in sodales dapibus, risus metus sagittis lorem, non porttitor purus odio nec odio. Sed tincidunt posuere elit. Quisque eu enim. Donec libero risus, feugiat ac, dapibus eget, posuere a, felis. Quisque vel lectus ut metus tincidunt eleifend. Duis ut pede. Duis velit erat, venenatis vitae, vulputate a, pharetra sit amet, est. Etiam fringilla faucibus augue.

5.2.1 Subsection One

Aenean velit sem, viverra eu, tempus id, rutrum id, mi. Nullam nec nibh. Proin ullamcorper, dolor in cursus tristique, eros augue tempor nibh, at gravida diam wisi at purus. Donec mattis ullamcorper tellus. Phasellus vel nulla. Praesent interdum, eros in sodales sollicitudin, nunc nulla pulvinar justo, a euismod eros sem nec nibh. Nullam sagittis dapibus lectus. Nullam eget ipsum eu tortor lobortis sodales. Etiam purus leo, pretium nec, feugiat non, ullamcorper vel, nibh. Sed vel elit et quam accumsan facilisis. Nunc leo. Suspendisse faucibus lacus.

5.2.2 Subsection Two

Pellentesque interdum sapien sed nulla. Proin tincidunt. Aliquam volutpat est vel massa. Sed dolor lacus, imperdiet non, ornare non, commodo eu, neque. Integer pretium semper justo. Proin risus. Nullam id quam. Nam neque. Duis vitae wisi ullamcorper diam congue ultricies. Quisque ligula. Mauris vehicula.

5.2.2.1 Subsubsection One

Curabitur nunc magna, posuere eget, venenatis eu, vehicula ac, velit. Aenean ornare, massa a accumsan pulvinar, quam lorem laoreet purus, eu sodales magna risus molestie lorem. Nunc erat velit, hendrerit quis, malesuada ut, aliquam vitae, wisi. Sed posuere. Suspendisse ipsum arcu, scelerisque nec, aliquam eu, molestie tincidunt, justo. Phasellus iaculis. Sed posuere lorem non ipsum. Pellentesque dapibus. Suspendisse quam libero, laoreet a, tincidunt eget, consequat at, est. Nullam ut lectus non enim consequat facilisis. Mauris leo. Quisque pede ligula, auctor vel, pellentesque vel, posuere id, turpis. Cras ipsum sem, cursus et, facilisis ut, tempus euismod, quam. Suspendisse tristique dolor eu orci. Mauris mattis. Aenean semper. Vivamus tortor magna, facilisis id, varius mattis, hendrerit in, justo. Integer purus.

5.2.2.2 Subsubsection Two

Vivamus adipiscing. Curabitur imperdiet tempus turpis. Vivamus sapien dolor, congue venenatis, euismod eget, porta rhoncus, magna. Proin condimentum pretium enim. Fusce fringilla, libero et venenatis facilisis, eros enim cursus arcu, vitae facilisis odio augue vitae orci. Aliquam varius nibh ut odio. Sed condimentum condimentum nunc. Pellentesque eget massa. Pellentesque quis mauris. Donec ut ligula ac pede pulvinar lobortis. Pellentesque euismod. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent elit. Ut laoreet ornare est. Phasellus gravida vulputate nulla. Donec sit amet arcu ut sem tempor malesuada. Praesent hendrerit augue in urna. Proin enim ante, ornare vel, consequat ut, blandit in, justo. Donec felis elit, dignissim sed, sagittis ut, ullamcorper a, nulla. Aenean pharetra vulputate odio.

5.2.2.2.1 Paragraph a Quisque enim. Proin velit neque, tristique eu, eleifend eget, vestibulum nec, lacus. Vivamus odio. Duis odio urna, vehicula in, elementum aliquam, aliquet laoreet, tellus. Sed velit. Sed vel mi ac elit aliquet interdum. Etiam sapien neque, convallis et, aliquet vel, auctor non, arcu. Aliquam suscipit aliquam lectus. Proin tincidunt magna sed wisi. Integer blandit lacus ut lorem. Sed luctus justo sed enim.

5.2.2.2.2 Paragraph b Morbi malesuada hendrerit dui. Nunc mauris leo, dapibus sit amet, vestibulum et, commodo id, est. Pellentesque purus. Pellentesque tristique, nunc ac pulvinar adipiscing, justo eros consequat lectus, sit amet posuere lectus neque vel augue. Cras consectetur libero ac eros. Ut eget massa. Fusce sit amet enim eleifend sem dictum auctor. In eget risus luctus wisi convallis pulvinar. Vivamus sapien risus, tempor in, viverra in, aliquet pellentesque, eros. Aliquam euismod libero a sem.

General Conclusion and Perspectives

Nunc velit augue, scelerisque dignissim, lobortis et, aliquam in, risus. In eu eros. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae; Curabitur vulputate elit viverra augue. Mauris fringilla, tortor sit amet malesuada mollis, sapien mi dapibus odio, ac imperdiet ligula enim eget nisl. Quisque vitae pede a pede aliquet suscipit. Phasellus tellus pede, viverra vestibulum, gravida id, laoreet in, justo. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Integer commodo luctus lectus. Mauris justo. Duis varius eros. Sed quam. Cras lacus eros, rutrum eget, varius quis, convallis iaculis, velit. Mauris imperdiet, metus at tristique venenatis, purus neque pellentesque mauris, a ultrices elit lacus nec tortor. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos hymenaeos. Praesent malesuada. Nam lacus lectus, auctor sit amet, malesuada vel, elementum eget, metus. Duis neque pede, facilisis eget, egestas elementum, nonummy id, neque.

Proin non sem. Donec nec erat. Proin libero. Aliquam viverra arcu. Donec vitae purus. Donec felis mi, semper id, scelerisque porta, sollicitudin sed, turpis. Nulla in urna. Integer varius wisi non elit. Etiam nec sem. Mauris consequat, risus nec congue condimentum, ligula ligula suscipit urna, vitae porta odio erat quis sapien. Proin luctus leo id erat. Etiam massa metus, accumsan pellentesque, sagittis sit amet, venenatis nec, mauris. Praesent urna eros, ornare nec, vulputate eget, cursus sed, justo. Phasellus nec lorem. Nullam ligula ligula, mollis sit amet, faucibus vel, eleifend ac, dui. Aliquam erat volutpat.



Future Perspectives. Fusce vehicula, tortor et gravida porttitor, metus nibh congue lorem, ut tempus purus mauris a pede. Integer tincidunt orci sit amet turpis. Aenean a metus. Aliquam vestibulum lobortis felis. Donec gravida. Sed sed urna. Mauris et orci. Integer ultrices feugiat ligula. Sed dignissim nibh a massa. Donec orci dui, tempor sed, tincidunt nonummy, viverra sit amet, turpis. Quisque lobortis. Proin venenatis tortor nec wisi. Vestibulum placerat. In hac habitasse platea dictumst. Aliquam porta mi quis risus. Donec sagittis luctus diam. Nam ipsum elit, imperdiet vitae, faucibus nec, fringilla eget, leo. Etiam quis dolor in sapien porttitor imperdiet.

Glossary

Term One Definition of the first term goes here.

Term Two Definition of the second term goes here.

Term Three Definition of the third term goes here.

Acronyms

Acronym	Full Form
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
DL	Deep Learning
IoT	Internet of Things

Supplementary Material

Cras pretium. Nulla malesuada ipsum ut libero. Suspendisse gravida hendrerit tellus. Maecenas quis lacus. Morbi fringilla. Vestibulum odio turpis, tempor vitae, scelerisque a, dictum non, massa. Praesent erat felis, porta sit amet, condimentum sit amet, placerat et, turpis. Praesent placerat lacus a enim. Vestibulum non eros. Ut congue. Donec tristique varius tortor. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Nam dictum dictum urna.

Phasellus vestibulum orci vel mauris. Fusce quam leo, adipiscing ac, pulvinar eget, molestie sit amet, erat. Sed diam. Suspendisse eros leo, tempus eget, dapibus sit amet, tempus eu, arcu. Vestibulum wisi metus, dapibus vel, luctus sit amet, condimentum quis, leo. Suspendisse molestie. Duis in ante. Ut sodales sem sit amet mauris. Suspendisse ornare pretium orci. Fusce tristique enim eget mi. Vestibulum eros elit, gravida ac, pharetra sed, lobortis in, massa. Proin at dolor. Duis accumsan accumsan pede. Nullam blandit elit in magna lacinia hendrerit. Ut nonummy luctus eros. Fusce eget tortor.

Bibliography

- [1] Atlassian. *Confluence: Team Workspace for Knowledge and Collaboration*. <https://www.atlassian.com/software/confluence>. Accessed: 2026. 2026.
- [2] Zoom Video Communications. *Zoom: Collaboration Tools and Video Meetings*. <https://www.zoom.com/>. Accessed: 2026. 2026.
- [3] Atlassian. *Jira: Project Management and Issue Tracking*. <https://www.atlassian.com/software/jira>. Accessed: 2026. 2026.
- [4] Slack Technologies. *Slack: Team Communication and Collaboration Platform*. <https://slack.com/>. Accessed: 2026. 2026.
- [5] Agile Alliance. *Principles Behind the Agile Manifesto*. <https://agilemanifesto.org/principles.html>. Accessed: 2026. 2001.